

M. Majestic Crossroads Al-Nasserah

time limit per test: 2 seconds
memory limit per test: 1024 megabytes

Al-Nasserah in the heart of Lower Galilee overlooks the Marj Ibn Amer plain and serves as a transition between this plain and the mountainous Upper Galilee. Historically significant, it connected main roads between Syria, Egypt, Jordan, and Palestine, attracting commercial caravans.

You are given three positive integers a, b, c . Your task is to count the number of divisors of $LD(a, b, c)$

Recall that the Ladallacian transform of any three integers a, b, c is defined as follows:

$$LD(a, b, c) = \sum_{i=0}^{(a+c)^b} (\lceil \frac{i}{a+c} \rceil - i\%(a+c))$$

Input

The input consists of three positive integers a, b, c ($1 \leq a, c \leq 10^9, 2 \leq b \leq 40$)

Additional constraint of the input: it is guaranteed that $LD(a, b, c) \leq 2^{100}$

Output

Output the number of divisors of $LD(a, b, c)$

Example

input	Copy
2 5 2	
output	Copy
22	

Note

Substituting with $a, b, c = 2, 5, 2$

$LD(2, 5, 2)$ will equal 130048 which has 22 divisors

ACPC Scientific Committee Archive

Private

Participant



About Group

https://www.youtube.com/@MoamenAhmed98/play

Group website

Group Contests

- ECPC Qualifications 2025 Day 3 (Mirror)
- ECPC Qualifications 2025 Day 2 (Mirror)
- ECPC Qualifications 2025 Day 1 (Mirror)
- TCPC 2025
- ACPC Girls 2025
- ACPC Girls 2020
- HPV-CPC 2021
- Tunisian Collegiate Programming Contest (2017)
- Azhar Collegiate Programming Contest 2019
- The 2018 Menofia University Collegiate Programming Contest
- ICPC International Collegiate Programming Contest, Assiut Collegiate Programming Contest (2019)
- International Collegiate Programming Contest, GUC Collegiate Programming Contest (2019)
- International Collegiate Programming Contest, Aleppo University Collegiate Programming Contest(2018)
- The 2025 ICPC Egyptian Collegiate Programming Contest for Teens(Online Qualification)
- ECPC Qualifications 2021 - Day 4
- ACPC Kickoff 2025
- International Collegiate Programming Contest, Al-Baath, Tishreen, Aleppo and Damascus Collegiate Programming Contest (2021)
- The 2020 Egyptian Collegiate Programming Qualification Contest Day 2
- OCPC 2025 (Qualification)
- OCPC 2025
- ACPC 2024 (Mirror)
- LCPC 2024 (Mirror)
- JCPC 2024 (Mirror)
- EthCPC (Mirror)
- PCPC 2024
- LyCPC 2024
- AOCPC 2024
- AOCPC 2023
- QCPC 2024 (Mirror)
- TCPC 2024 (Mirror)